Compatibility Findings: Classical Mechanics and FakeScience

Formalization notes

November 12, 2026

1 Findings

The two supplied Lean files are not jointly importable as written. Both declare the fully qualified name `ClassicalMechanics.secondLawHolds`, but give it different types. In `ClassicalMechanics.lean`, it states, for a system M and a particle p at time t,

$$\text{forceSum}(M.\text{force}(p,t)) = \text{scale}(M.\text{mass}(p), M.\text{acceleration}(p,t)).$$

This extracts Newton's second law from the system's assumptions; it is the P_6 clause of the axiomatic presentation [1]. In `FakeScience.lean`, the same name is instead a theorem of type `False`. Lean rejects both a joint import and the combined declarations because the name is already declared. This diagnostic is a *name collision*, not a theorem that the two physical claims contradict each other.

The fake file also declares

```
axiom ClassicalMechanics.physics :  False.
```

This is a different and deeper problem: the file assumes a proof of `False`. From that assumption Lean's `False.elim` derives any proposition Q, so the resulting theory is logically inconsistent relative to ordinary Lean logic. Lean accepts the file because it permits users to introduce axioms; acceptance does not validate those axioms. The fake theorem explicitly depends on `physics`; the genuine `secondLawHolds` has no declared axiom dependencies. The supplied fake file does not state an alternative force law or supply a model, so no specific physical conflict with P_6 can be inferred from it. Renaming the duplicate theorem would resolve the name collision, but would not repair the false axiom.

2 Reproduction

With Lean 4 installed, run from the repository root:

```
bash _lean/check_compatibility.sh
```

The script compiles both files separately, checks the axiom dependencies, and verifies that joint import and combined declarations fail with the expected name-collision diagnostics. It also checks `FakeScienceAudit.lean`, which derives an arbitrary proposition from the fake axiom and reports the dependency using `#print axioms`. The checks reported here were run with Lean 4.35.0; the script checks specific diagnostic wording, which may change between Lean versions. On a machine using `elan` without a default toolchain, set `ELAN_TOOLCHAIN` to an installed Lean 4 toolchain first.

These checks do not prove that the abstract classical mechanics signature has a concrete real-analysis interpretation, nor do they check the truth of any empirical paper. They locate the collision and show exactly how the fake file's explicit assumption makes arbitrary propositions derivable.

References

[1] J. C. C. McKinsey, A. C. Sugar, and Patrick Suppes. Axiomatic foundations of classical particle mechanics. *Journal of Rational Mechanics and Analysis*, 2(2):253–272, 1953.